Sébastien Naze, PhD

COMPUTATIONAL NEUROSCIENTIST

sebastien.naze@gmail.com

+1 332-254-5090

55 Loisaida Avenue
Manhattan, New York

French citizen
U.S. Green card holder

"The best way to make your dreams come true is to wake up."

Paul Valéry

PROGRAMMING SKILLS

Python, C/C++, Java
Distributed and parallel computing (MPI, DASK)
Multi-Objective Optimization (MCMC, SMC)
Object Oriented & Service Architectures
Cloud Computing (docker, AWS, SQLite, Redis)
Tensor computing (PyTorch, Einops)
Git – GitLab
Jupyter Lab / Data Eng. (Pandas, JAX, …)
Agile methodology – CI/CD

SCIENTIFIC SKILLS

Time series analysis (Fourier, Wavelets)
Machine Learning (classical, LLMs, RL, EBMs)
Network Science (e.g. Graph Signal Processing)
Dynamical Systems Modeling & Analysis
Neuroimaging & Bayesian Methods

COMMUNICATION & LEADERSHIP

Client presentations & Tutorials
Student / Intern Supervision
Fluent in English, French & Spanish

PROFESSIONAL EXPERIENCE

QIMR RESEARCH FELLOW

Queensland Institute of Medical Research, Brisbane, Australia *2021 - current*

- Data engineering, analysis of multimodal dataset, multiscale modeling, and funding acquisition for neurostimulation in obsessive-compulsive disorders.
- Development of Neuro-AI pipeline for clinical prediction and mechanistic interpretability.

IBM RESEARCH SCIENTIST

IBM Research, Melbourne, Australia *2019 - 2021*

- Designed and implemented an interface between brain modeling and machine learning algorithms in Python and C++.
- Performed parameter optimization using evolutionary algorithms.
- Integrated libraries into service architecture for cloud computing.
- Worked in international team of 4-12 people operating from multiple countries.

IBM POSTDOCTORAL RESEARCHER

IBM T. J. Watson Research Center, New York, USA *2016 – 2019*

- Delivered IBM Solutions in brain modeling to clients (CHDI, Pfizer) investigating Huntington's disease phenotypes in-silico.
- Delivered quarterly reports, client presentations and software patches.
- Served as team co-lead on collaborative research projects with academic partners (Stanford, Brown University, Univ. of Oxford, UHN Toronto).

RESEARCH ASSISTANT

Institute of Systems Neuroscience, Marseille, France *2011 – 2015*

Preparation of PhD Thesis entitled Multiscale Modeling of Epileptic Seizure Dynamics.

TECHNICAL ASSISTANT

Vrije Universiteit, Amsterdam, The Netherlands *2010 – 2011*

Preparation of MSc Thesis entitled Computational Agent Modeling of Post-Traumatic Stress Disorders.

EDUCATION

PHD COMPUTATIONAL NEUROSCIENCE

Aix-Marseille Université, France
2015

MSC INFORMATION SCIENCE

Vrije Universiteit, The Netherlands
2011

BSC SOFTWARE ENGINEERING

Heriot-Watt University, United Kingdom
2009

PUBLICATIONS



Naze S, et al. (2025) Mechanisms and interventions promoting healthy frontostriatal dynamics in obsessive-compulsive disorder. *Nature Communications.*

Webb L., *(…),* **Naze S**., Cocchi L. (2025) Altered striatal functional gradients in obsessive-compulsive disorder. *Biological Psychiatry.*

L. Cocchi, **Naze S**, et al. (2023) Effects of noninvasive stimulation of the rostromedial prefrontal cortex on symptoms and frontostriatal connectivity in obsessive-compulsive disorder. *Nature Mental Health*.

Naze S, et al. (2023) Mechanisms of imbalanced frontostriatal functional connectivity in obsessive-compulsive disorder. *Brain.*

Sihag S, **Naze S**, et al. (2022) Functional brain activity constrained by structural connectivity reveals cohort-specific features for serum neurofilament light chain. *Nature Communications Medicine*.

Naze S, et al. (2021) Features importance in seizure classification using scalp EEG reduced to single timeseries. *IEEE EMBC.*

Naze S, et al. (2021) Robustness of connectome harmonics to local gray matter and long-range white matter connectivity changes. *Neuroimage*.

Sihag S, **Naze S**, et al. (2020) Multimodal Dynamic Brain Connectivity Analysis Based on Graph Signal Processing for Former Athletes With History of Multiple Concussions. *IEEE Transactions on Signal and Information Processing over Networks.*

Sihag S, **Naze S**, et al. (2019) GSP Analysis of Brain Imaging Data from Athletes with History of Multiple Concussions. *IEEE ISIP.*

Naze S, et al. (2019) Classification of TMS evoked potentials using ERP time signatures and SVM versus deep learning. *IEEE EMBC.*

Naze S, et al. (2018) Cortico-striatal cross-frequency coupling and gamma genesis disruptions in Huntington's disease mouse and computational models. *eNeuro 5:6*.

Naze S, et al. (2015) Computational modeling of seizure dynamics using coupled neuronal networks: factors shaping epileptiform activity. PLoS Comp Biol 11:5.

Bernard C, **Naze S**, et al. (2014) Modern concepts of seizure modeling. *Annual Reviews on Neurobiology* 114:120-153.

Naze S, Treur J. (2012) A Computational Agent Model for Post-Traumatic Stress Disorders. *International Conference on Neural Information Processing*.

Naze S, Treur J. (2011) A computational model for development of post-traumatic stress disorders by hebbian learning. *Biologically-inspired Cognitive Architectures.*

CONFERENCE PRESENTATIONS

Naze S. (2025) Frontostriatal dynamics modelling obsessions and compulsions. *Cognitive Computational Neuroscience (CCN),* Amsterdam, The Netherlands.

Naze S. (2025) Mechanisms and interventions promoting healthy frontostriatal dynamics in obsessive-compulsive disorder. *Organisation for Human Brain Mapping (OHBM),* Brisbane, Australia.

Naze S. (2022) Mechanisms of imbalanced fronto-striatal functional connectivity in obsessive-compulsive disorder. *Biological Psychiatry Australia,* Newcastle, Australia.

Naze S. (2021). Features importance in seizure classification using scalp EEG reduced to single timeseries. *IEEE Engineering in Medicine and Biology Conference* (Online).

Naze S. (2020). Functional requirements of intentional control over the integrated cortico-thalamo-cortical and basal ganglia systems using neural computations. *Organization for Computational Neurosciences 29th Annual Meeting* (Online).

Naze S. (2019) Classification of TMS evoked potentials using ERP time signatures and SVM versus deep learning. *IEEE Engineering in Medicine and Biology Conference,* Berlin, Germany.

Naze S. (2019) Robustness of connectome harmonics to local gray matter and long-range white matter connectivity changes. *Organization for Computational Neurosciences 28th Annual Meeting,* Barcelona, Spain.

Naze S. (2018) Sensitivity analysis of the connectome harmonics framework and applications to neurodegenerative diseases. *Organization for Computational Neurosciences 27th Annual Meeting,* Seattle, United Sates.

Naze S. (2017) Cortico-striatal cross-frequency coupling and gamma genesis disruptions in Huntington's disease mouse and computational models. *Society for Neuroscience,* Washington D.C., United States.

Naze S. (2015) Computational modeling of initiation and propagation of epileptic seizures. *7th International Workshop on Seizure Prediction,* Melbourne, Australia.

Naze S. (2011) A Computational Agent Model of Post-Traumatic Disorders. *Biologically-Inspired Cognitive Architectures,* Washington D.C., United States.

AWARDS AND GRANTS

2023 National Health and Medical Research Council of Australia (NHMRC) IDEAS Grant number 2027597:
"Noninvasive ultrasonic deep brain stimulation as a novel tool to change brain activity in obsessive-compulsive disorder"
as Chief Investigator (CIC) ***1.246.630 $AU***

REFERENCES

Dr. James Kozloski (Manager)
IBM Research, 1101 Kitchawan Road, Yorktown Height, New York 10598, United States.
kozloski@us.ibm.com

Dr. Luca Cocchi (Principal Investigator)
QIMR Berghofer Medical Research Institute, 300 Herston Road, Brisbane 4006, Australia.
luca.cocchi@qimrberghofer.edu.au

Dr. Stefan Harrer (Chief Innovation Officer)
Digital Health CRC, 520 Bourke Street, Melbourne 3000, Australia.
stefan.harrer@dhcrc.com

Dr. Viktor Jirsa (Director)
Institut de Neurosciences des Systèmes, 27 Boulevard Jean Moulin, 13005 Marseille, France.
viktor.jirsa@univ-amu.fr

Dr. Christophe Bernard (Team head)
Institut de Neurosciences des Systèmes, 27 Boulevard Jean Moulin, 13005 Marseille, France.
christophe.bernard@univ-amu.fr